Oberseminar Data Science

Topic Presentation — Summer Semester 2026

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University of Passau · Chair of Data Science

What is the Oberseminar?

The Oberseminar is the entry point for **Bachelor's and Master's theses** at the Chair of Data Science.

Format

- Presentation and Q&A for students conducting a thesis at the chair
- Topic presentation: twice a semester
- In-between sessions: students presenting their topic, their final presentation and Q&A on the thesis.

Who attends

- Students looking for a BSc or MSc thesis topic
- Students currently working on their thesis (progress updates)

What today covers

- The registration process — step by step
- What we expect from you
- Defined topics (overview + link to full briefs)
- The open theme and how to apply

No gos

- No thesis topics with companies
- No thesis topic outside of the suggested topics (except for excellent, research tier topics)
- No thesis topics that are not related to the research topics of the chair

> Leave today with one topic in mind — and know exactly what to do next.

Thesis Process

Step 1 — Topic Selection

Pick one topic from the defined list, or prepare a one-pager for the open theme. Ask questions now. No guarantees you are getting a topic. We will select best on affinity to the chair (courses, grades).

Step 2 — Entry Task

A 1–2 week scoped task assigned with the topic. It tests whether you and the topic are a good match.

Step 3 — Research Exposé

A structured 4–6 page document (motivation, state of the art, research questions, methodology, work plan, limitations). Presented at the end-of-semester Oberseminar.

Step 4 — Registration

Once the exposé is accepted, register with the examination office. **Registration is binding** — it starts the clock.

```
Topic presentation
        ↓
  Entry task
  (1-2 weeks)
        ↓
 Research exposé
  (4-6 pages)
        ↓
Exposé presentation
  (Oberseminar)
        ↓
 Official registration
        ↓
  6-month working
    period (MSc)
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You **cannot** register the thesis until the exposé has been accepted. A Master's thesis must be **submitted within six months** of registration; Bachelor's theses have a shorter period per the examination regulations.

What We Expect

Scientific standard

- Clear research question — not a project description
- Related work that goes beyond tutorial summaries
- Reproducible experiments with baselines and metrics
- Honest limitations and scope boundaries

Working style

- Self-directed — you drive the schedule
- Weekly or biweekly check-ins with your supervisor
- All code and data in a Git repository from day one
- Written communication in English

The exposé — required structure

Section	Content
Motivation	Why this problem matters now
State of the art	What exists and where it falls short
Research questions	At least 3 specific, answerable questions
Methodology	Datasets, metrics, tools, evaluation plan
Work plan	3–5 milestones with realistic dates
Limitations	Known constraints and threats to validity

Limitations are mandatory. A research plan without stated limitations signals insufficient understanding of the problem scope.

Meeting Prof. Granitzer

Weekly Q&A slot

- **Monday, 14:00–16:00** - presentation slots as defined in Stud.IP Abschlussarbeiten. You should be here to listen.
- Requires **prior registration through the chair's secretariat** — drop-ins are not possible
- Bring one concrete question or a short one-pager; unfocused check-ins get deferred
- For the open OUR²S / YOARS theme, align on scope in the Q&A **before** submitting the one-pager

Contact

- Secretariat — to book the Q&A slot
- Email — michael.granitzer@uni-passau.de

Where to find the full topic briefs

All defined topics live on the chair's master thesis-topics page. The slides that follow are pitch cards — the full briefs (research questions, methodology, why-it-fits) are maintained there and are the source of truth.

- Master page: /research/thesis-topics/
- How to apply: /research/thesis-topics/overview/

> Read the master brief before the Q&A. The slot is for alignment and scope, not for retelling the brief.

Defined Topics

Pitch Cards — Full Briefs on the Master Page

> Each topic comes with a 1–2 week entry task. Completing it well is the condition for advancing to the exposé phase. Full research questions, methodology, and background live on the master thesis-topics page.

Topic 1: Fast URL Risk Classifier

Scope: Bachelor's thesis or compact Master's project thesis

Pitch: A URL-only classifier that flags spam, ads, and security-risk URLs in microseconds — lightweight enough to run at web scale, data-driven enough to generalise beyond hand-written domain rules.

Connects to: OUR²S pipeline — fast URL-level filtering is a practical bottleneck before full document retrieval.

Full brief: /research/thesis-topics/#url-risk-classifier

Speaker notes

Entry task: Given a small labelled URL dataset, implement a character n-gram feature extractor and train a logistic regression classifier. Report precision, recall, and inference time per URL.

Topic 2: Ranking Profiles with Client-Side XGBoost

Scope: Master's thesis or research prototype

Pitch: Expose a small family of named ranking profiles (e.g. "freshness-first", "authority-first", "diversity-first") that clients select at query time. Discover them without explicit preference labels. Deployable as a real OUR²S API feature.

Connects to: OUR²S search API — directly extends the chair's infrastructure.

Full brief: /research/thesis-topics/#ourrs-ranking-profiles

Speaker notes

Entry task: Using a provided OUR²S query log sample, cluster queries by result-feature distributions and identify two or three clearly distinct ranking biases. Describe each in plain language.

Topic 3: Hierarchical Multilabel Classification for OpenWebIndex

Scope: Master's thesis

Pitch: Replace the current rule-based category assignment on the Open Web Index with a data-driven hierarchical multilabel classifier over the Curlie taxonomy. Compare Jina v5 embeddings against TF-IDF / BM25 baselines on both accuracy and throughput.

Connects to: Open Web Index — the dataset produced would be a public research contribution.

Full brief: /research/thesis-topics/#openwebindex-hierarchical-classification

Speaker notes

Entry task: Given 1000 labelled web pages from the Curlie dataset, train a TF-IDF logistic regression classifier and report precision@1 and precision@3 on a held-out set. Describe where it fails.

Topic 4: Multi-Objective Alignment for LLM Fine-Tuning

Scope: Master's thesis

Pitch: Alignment is usually optimised as one scalar reward, but helpfulness, factuality, safety, and task success genuinely conflict. Map the Pareto front; evaluate whether post-hoc profile switching can serve multiple alignment regimes from one base model.

Connects to: Agentic AI — aligned agents must balance competing objectives in real deployments.

Full brief: /research/thesis-topics/#multi-objective-alignment

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Speaker notes

Entry task: Using a small instruction-tuned model and two alignment metrics (e.g. helpfulness and factuality), run fine-tuning at three different objective weightings and report whether the trade-off is observable.

Topic 5: REST API Design and Control Primitives for AI Agent Systems

Scope: Master's thesis

Pitch: Re-evaluate classic REST principles — HATEOAS, modularity, DRY, idempotency, least privilege, self-description — empirically for LLM agents that plan across calls. Extend with agent-control primitives (permission scoping, interruption hooks, state-transition constraints) that reduce unsafe or unrecoverable actions.

Connects to: OUR²S API and agentic search — benchmark spans both REST design and control-primitive angles.

Full brief: /research/thesis-topics/#rest-api-design-for-agents

Speaker notes

Entry task: Take a small REST API and add HATEOAS links to all responses. Run an LLM agent on two tasks: one it has seen before, one it has not. Compare completion rates and error types with and without HATEOAS. As a second step, add an explicit state machine and compare agent recovery behaviour.

Topic 6: Minimal LLM Agents for API-Specific Tasks

Scope: Master's thesis

Pitch: General-purpose LLMs are expensive and brittle for API-bounded agent tasks. Study whether small fine-tuned models can reliably adhere to a specific API — and how minimal they can be while staying reliable on structured, API-specific workflows.

Connects to: Complements Topic 5 — same benchmark APIs, different angle: model optimisation vs. API design.

Full brief: /research/thesis-topics/#minimal-llm-agents-api

Speaker notes

Entry task: Fine-tune a small instruction-tuned model (1–3B parameters) on a synthetic dataset of correct API calls for one simple REST API. Compare completion rate and hallucinated-endpoint rate against a larger base model without fine-tuning.

Open Theme

Apply with a One-Pager

Open themes have no fixed specification. You define the research direction within the theme; selection is based on the quality of your one-page proposal.

Open Theme: Web Search with OUR^{2}S / YOARS

Context

The Chair of Data Science runs its own open-web search infrastructure built on the Open Web Index. Unlike commercial search engines, the system is fully transparent: retrieval behaviour, ranking, and result provenance can be inspected, compared, and modified. **OUR^{2}S** (Open Web Index Retrieval Research System) and **YOARS** (Your Open Web Index Agentic Retrieval Search System) provide the infrastructure for experiments on real web-scale data.

Why this is open

Scope

Bachelor's thesis or Master's thesis

How to apply

Submit a **one-page proposal** by email before the entry-task phase. The one-pager must include:

1. A concrete research question (not just a topic area)
2. Why you are the right person to work on it (background, skills, motivation)
3. A plausible evaluation strategy

Selection is based on proposal quality. A well-scoped question with a credible evaluation plan outweighs domain expertise alone.

The research question — and therefore the thesis — is yours to define within the theme of RAG, agentic search, or web retrieval.

Example directions

- When does query reformulation improve agentic search on exploratory tasks?
- How do sparse, dense, and hybrid retrieval differ on navigational vs. informational queries?
- Can retrieval provenance improve LLM answer faithfulness?
- How does document freshness interact with ranking quality?

Infrastructure access

- Search API and data: ourrs.eu/developers
- Login via B2ACCESS with University of Passau credentials
- Corpus: Open Web Search Curlie 2025

> Align on scope with Prof. Granitzer in the Monday Q&A slot **before** writing the one-pager.

Next Steps

Defined topics (Topics 1–6)

1. Pick one topic and read its full brief on the master page
2. Book the Q&A slot through the secretariat (Monday 14:00–16:00)
3. Express interest; receive the entry task
4. Complete the entry task (1–2 weeks)
5. If accepted: write the research exposé
6. Present the exposé at the end-of-semester Oberseminar
7. Register the thesis (starts the 6-month MSc clock)

Master page: /research/thesis-topics/

Contact: michael.granitzer@uni-passau.de

Open theme (OUR²S / YOARS)

1. Read the OUR²S infrastructure docs at ourrs.eu
2. Develop a concrete research question
3. Submit the one-pager (you have one try)
4. Selection is based on proposal quality and past performance at lectures at the chair.

> One-pagers that describe a topic area without a concrete research question will not be considered.

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One topic, with proper motivation, research gap, a core research question and expected outcome. Be concrete and concise.